

EXP:3.1 DECISION TREE CLASSIFICATION

AIM:

To implement Decision Tree classification using the Iris dataset and evaluate the performance of the model using accuracy score and confusion matrix.

SOFTWARE REQUIREMENTS:

- Python 3.x
- Jupyter Notebook
- NumPy
- Pandas
- Scikit-learn
- Matplotlib
- Seaborn

DATASET:

Iris Dataset

The Iris dataset contains 150 samples belonging to three classes:

1. Setosa
2. Versicolor
3. Virginica

The four input features are:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

ALGORITHM:

1. Import the required Python libraries.
2. Load the Iris dataset from Scikit-learn.
3. Separate the dataset into input features X and target variable y.
4. Split the dataset into training and testing data.
5. Create a Decision Tree Classifier using the **Gini criterion**.
6. Train the model using the training data.
7. Predict the classes for the test data.

8. Calculate the confusion matrix.
9. Calculate the accuracy score.
10. Visualize the confusion matrix.
11. Visualize the trained decision tree.

CODE

```
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import confusion_matrix, accuracy_score

iris = load_iris()

X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = DecisionTreeClassifier(
    criterion="gini",
    random_state=0
)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

cm = confusion_matrix(y_test, y_pred)

accuracy = accuracy_score(y_test, y_pred)

print("Confusion Matrix:")
print(cm)

print("\nAccuracy Score:", accuracy)

plt.figure(figsize=(6, 5))

sns.heatmap(
```

```

cm,
annot=True,
fmt="d",
cmap="Blues",
xticklabels=iris.target_names,
yticklabels=iris.target_names
)

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix - Gini Index")
plt.show()

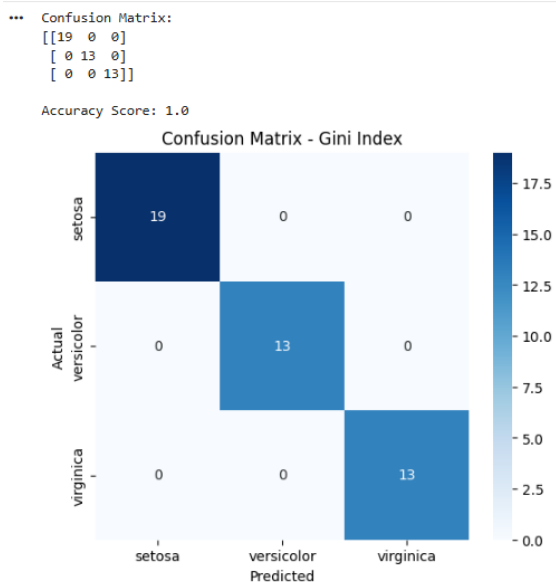
plt.figure(figsize=(15, 8))

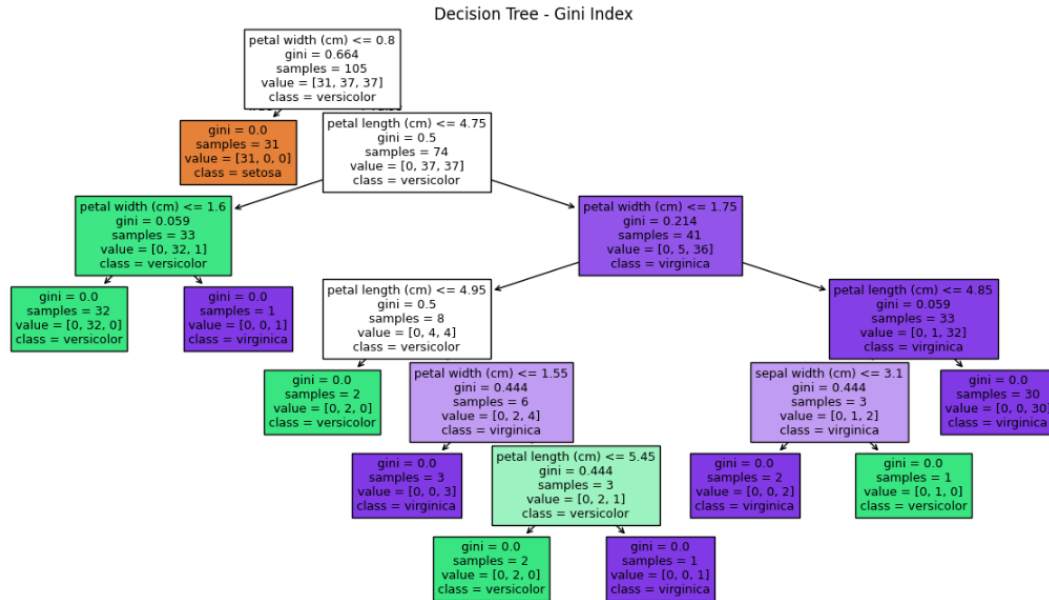
plot_tree(
    model,
    filled=True,
    feature_names=iris.feature_names,
    class_names=iris.target_names
)

plt.title("Decision Tree - Gini Index")
plt.show()

```

OUTPUT:





RESULT:

The Decision Tree classifier using the **Gini Index** was successfully implemented using the Iris dataset. The model was trained and tested successfully, and its performance was evaluated using the confusion matrix and accuracy score. The decision tree was also visualized to understand the classification process.